



MODELING A NIFE CATALYST FOR DRY METHANE REFORMING WITH DENSITY FUNCTIONAL THEORY

COMPUTATIONAL MATERIALS SCIENCE

TOOLS & METHODS:

Quantum ESPRESSO (pw.x geometry relaxation, neb.x nudged elastic band, cp.x) · Avogadro structure construction · Materials Project and CCDC reference structures · MPI-parallel execution on 64 cores with -npool / -ndiag tuning · post-processing and energy-profile analysis in Excel.

EXECUTIVE SUMMARY:

Nearly all industrial hydrogen comes from steam methane reforming at 700–1000 °C, a process that emits CO₂. Dry reforming of methane (DRM) consumes CO₂ as the oxidant instead, but it is strongly endothermic and its catalysts either cost too much (noble metals) or deactivate through carbon deposition. Bimetallic Ni–Fe on oxide supports has been reported to resist coking and remain active at lower temperatures.

This study used density functional theory with the nudged-elastic-band method to map the DRM minimum-energy pathway with and without a NiFe catalyst. The uncatalyzed pathway shows the expected large barrier; the catalyzed pathway converged only partially within the compute budget, but every converged image lay lower in energy, consistent with favorable methane chemisorption. Results are reported with explicit limits on what that partial convergence establishes.

KEY RESULTS:

- **Uncatalyzed pathway (26 images):** barrier ≈ 1.1 Ry with a broad transition region; products ~ 0.3 Ry above reactants, so the reaction is strongly endothermic.
- **Catalyzed pathway:** 5 of 26 images converged; energy fell monotonically from -6943.6 to -6998.8 Ry through the chemisorption steps.
- **Mechanism resolved:** CH₄ $\sigma^*(\text{C-H})$ orbitals mix with NiFe d-states to weaken C–H bonds; support lattice oxygen forms CO and CO₂ refills the vacancy.
- **Barrier width explained** by the sequential breaking of four sp³ C–H bonds and one C=O double bond.
- **Method selection validated:** ab-initio molecular dynamics abandoned in favour of NEB once it proved unable to resolve a transition state.
- **Convergence gap quantified** with a costed plan to close it on GPU or dedicated compute nodes.

INTRODUCTION:

Hydrogen is central to decarbonizing steel, ammonia and heavy transport, yet producing it from methane today releases roughly nine kilograms of CO₂ per kilogram of H₂. Dry reforming turns two greenhouse gases into syngas (2 H₂ + 2 CO), but its ~ 1000 °C operating point and rapid catalyst coking have kept it out of industrial service.

The catalyst is the constraint. Noble metals such as platinum resist coking but are prohibitively expensive at scale; nickel is inexpensive but deactivates within hours as carbon accumulates on the active surface. Ni–Fe nanoparticles grown on perovskite supports have been shown experimentally to suppress coking and increase activity, which raises the question this study addresses: does such a surface meaningfully lower the reaction barrier, and through what mechanism?

TEST SETUP:

Method selection. Mapping a reaction pathway by hand would require constructing roughly 24 intermediate geometries and running a self-consistent-field calculation on each. The nudged-elastic-band method instead takes the reactant and product, plus optional intermediates, then interpolates the images automatically, and relaxes the whole band toward the minimum-energy path while holding the images evenly spaced. Because the target quantity was an activation barrier, 0 K electronic energies were judged sufficient; temperature-dependent NEB across multiple temperatures fell outside the available compute budget.

Model systems. Two systems were constructed in Avogadro: an isolated CH₄ + CO₂ pair as the gas-phase reference, and the same reactants positioned above a NiFe slab on a lanthanum-oxide (perovskite-type) support. The gas-phase geometry was relaxed with pw.x prior to the NEB run. For the catalyst system, curated structures from the Materials Project and the CCDC replaced an in-house relaxation to conserve compute. Each NEB was configured with 26 images, and the catalyst band included two manually constructed intermediates, dissociated CH₄ with H₂ released and CO formed from lattice oxygen, to guide the path.

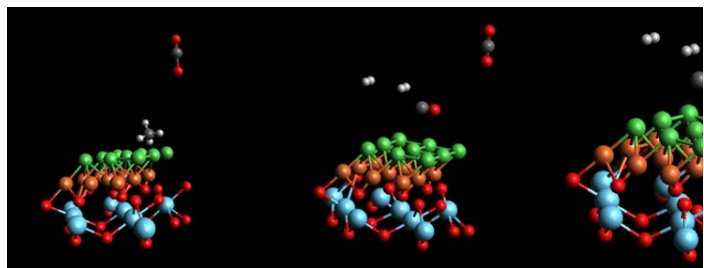


FIG 1: Images supplied to the catalyst NEB, left to right: CH₄ and CO₂ approaching the NiFe surface (green Ni, orange Fe) over the oxide support; H₂ released as methane dissociates; CO formed from lattice oxygen; final 2 H₂ + 2 CO with the vacancy refilled by CO₂.

Execution. Each job ran on 64 processors of a shared node under a 20-hour wall-time limit, using -npool 32 and -ndiag 32 for the larger catalyst system. The gas-phase band converged fully; the catalyst band, carrying an order of magnitude more atoms, converged five images within the limit.

RESULTS:

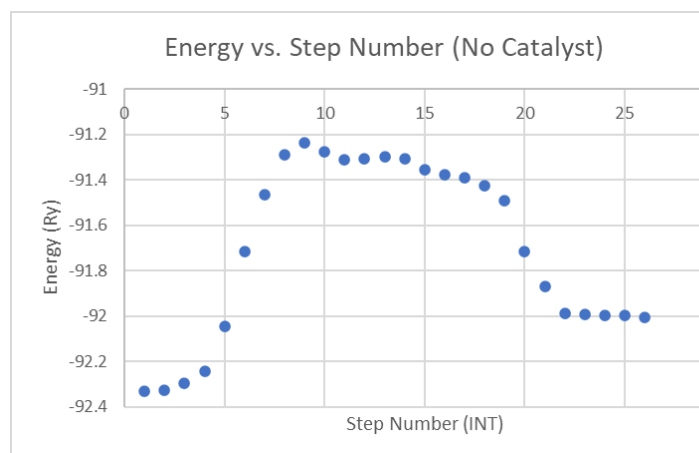


FIG 2: Total energy versus NEB image for the uncatalyzed reaction: a ~1.1 Ry climb to a broad plateau, then relaxation to products that still sit above the reactants.

In the gas phase the energy rises from -92.3 Ry to a plateau near -91.2 Ry and settles at -92.0 Ry for the products: a barrier of about 1.1 Ry and an endothermic net change. The breadth of the barrier reflects the sequential breaking of four stable sp³ C-H bonds and a C=O double bond rather than a single concerted step.

On the NiFe surface the five converged images fall from -6943.6 to -6998.8 Ry. Physically this corresponds to methane chemisorbing on the metal: its antibonding C-H orbitals mix with Ni and Fe d-orbitals, electron density between carbon and hydrogen is depleted, and dissociation becomes energetically downhill. Lattice oxygen from the support then oxidizes the surface carbon to CO, and CO₂ heals the resulting vacancy and releases the second CO. This is a Mars-van Krevelen-type cycle in which the support participates directly rather than acting as an inert scaffold.

WHAT THE RESULT DOES, AND DOES NOT, ESTABLISH

The catalyzed profile demonstrates a strongly favorable adsorption and dissociation step, not a validated activation barrier: with 21 images unconverged, no transition state was located. The slab was also not relaxed, and it exposed the (100) facet rather than the more reactive (111). The value of the study lies in the method, the mechanism it makes explicit, and a costed plan for closing the gap.

APPLICATIONS:

Dry reforming is one of the few routes that consumes carbon dioxide while producing a commodity chemical, which makes catalyst screening directly relevant to hydrogen producers, ammonia and methanol plants, and operators of biogas or landfill-gas streams already rich in both CH₄ and CO₂. Replacing platinum-group metals with nickel-iron would remove the single largest materials cost from such a process.

The computational workflow demonstrated here transfers beyond this reaction. Any surface-catalyzed process where the question is "which candidate surface lowers the barrier, and why" can be screened this way before committing to synthesis and reactor time, at a small fraction of the cost of an experimental campaign. The practical requirements for a production-grade screen are known: programmatic structure generation for reproducible (111) slabs, charge-density post-processing to locate the active site before the pathway is computed, staged coarse-then-refined bands to manage convergence, and GPU or dedicated compute nodes rather than shared CPU allocation.

DISCUSS THIS WORK:

This paper summarises one project. If the same problem appears in your programme, I am happy to walk through the method, the data behind it, and how it would map onto your hardware. emincherif@gmail.com